



PicoMagnetometer RPI Python code install manual

Pico^{nT}

<https://ukraa.com/>

The UK Radio Astronomy Association

A charitable incorporated organisation

Registered charity in England and Wales No. 1123866



Python code for the UKRAA PicoMagnetometer

Set of Python code to run on an RPi4/5 to get, process and present data from the UKRAA PicoMagnetometer

This software was written to suit a specific set-up, feel free to use as you see fit.

Instructions for setting up a Raspberry Pi4/5 are included in the **docs** folder

Table of Contents

Python code for the UKRAA PicoMagnetometer	2
Using the code	4
Where is my PicoMagnetometer?	5
Getting the software onto your RPi	6
Install the software onto your RPi	7
What does the code do?	9
Check GetDataRaw.py is running	10
License	11
Contact us	11

Using the code

The code assumes that your UKRAA PicoMagnetometer is connected to the RPi4/5 via supplied USB cable and that it is **/dev/ttyACM0** - you can check this by using **ls /dev/tty*** in a terminal window on the RPi4/5 and reviewing the response.

The code assumes username is **pi**.

If **pi** is not the username, then you will need to change all occurrences of **'/home/pi'** to **'/home/username'** in the python and gnuplot scripts, where **username** is the username you have selected for your RPi4/5.

The code assumes one magnetometer connected to the RPi4/5 USB and that it will be connected via **/dev/ttyACM0**.

If there are other devices connected to the RPi and your magnetometer is not **/dev/ttyACM0**, then you will need to change **/dev/ttyACM0** to **/dev/ttyACMx** in the **GetDataRaw.py** python script, where **ttyACMx** is the tty address of you connected magnetometer.

GetDataRaw.py is run as a service.

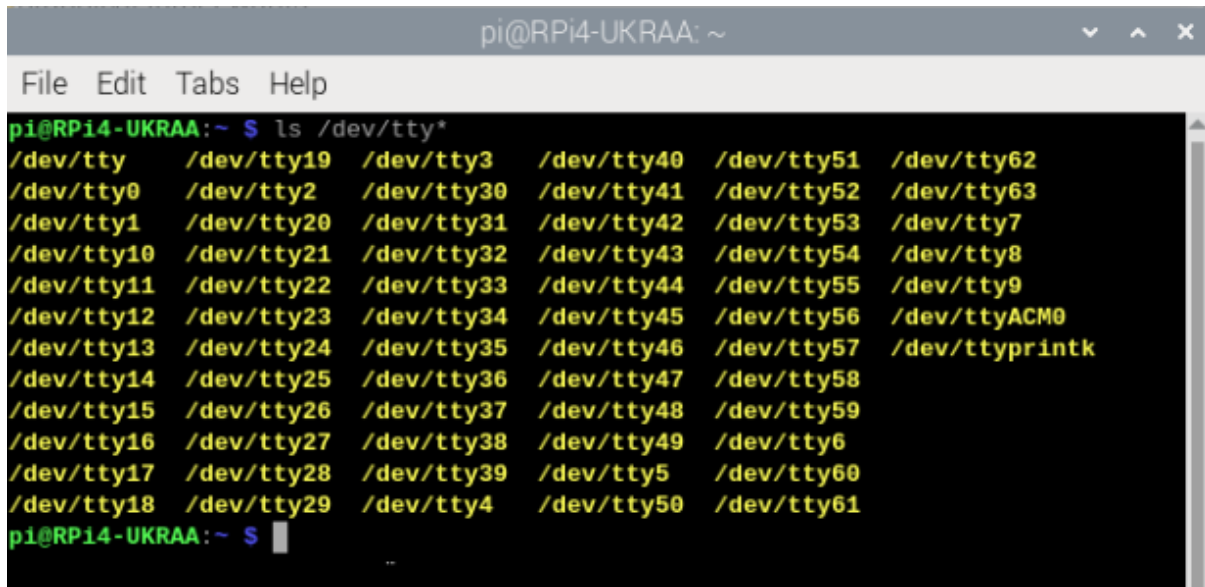
Other scripts (Python, gnuplot and shell) are run from **cron**.

Where is my PicoMagnetometer?

Plug your PicoMagnetometer into any of the RPi USB ports - I normally use one of the blue ports (USB3).

1. Open a terminal window and type the following command and press enter

ls /dev/tty*



```
pi@RPi4-UKRAA: ~  
File Edit Tabs Help  
pi@RPi4-UKRAA:~$ ls /dev/tty*  
/dev/tty /dev/tty19 /dev/tty3 /dev/tty40 /dev/tty51 /dev/tty62  
/dev/tty0 /dev/tty2 /dev/tty30 /dev/tty41 /dev/tty52 /dev/tty63  
/dev/tty1 /dev/tty20 /dev/tty31 /dev/tty42 /dev/tty53 /dev/tty7  
/dev/tty10 /dev/tty21 /dev/tty32 /dev/tty43 /dev/tty54 /dev/tty8  
/dev/tty11 /dev/tty22 /dev/tty33 /dev/tty44 /dev/tty55 /dev/tty9  
/dev/tty12 /dev/tty23 /dev/tty34 /dev/tty45 /dev/tty56 /dev/ttyACM0  
/dev/tty13 /dev/tty24 /dev/tty35 /dev/tty46 /dev/tty57 /dev/ttyprintk  
/dev/tty14 /dev/tty25 /dev/tty36 /dev/tty47 /dev/tty58  
/dev/tty15 /dev/tty26 /dev/tty37 /dev/tty48 /dev/tty59  
/dev/tty16 /dev/tty27 /dev/tty38 /dev/tty49 /dev/tty6  
/dev/tty17 /dev/tty28 /dev/tty39 /dev/tty5 /dev/tty60  
/dev/tty18 /dev/tty29 /dev/tty4 /dev/tty50 /dev/tty61  
pi@RPi4-UKRAA:~$
```

2. You are looking for **/dev/ttyACM0** - this is on the right hand side of the screen shot above.
3. This is the USB address for your attached PicoMagnetometer - if you have more than one magnetometer attached you may see **/dev/ttyACM1** etc.
4. If you do not see **/dev/ttyACM0**, then unplug and plug the PicoMagnetometer back in and try again.
5. As long as you see **/dev/ttyACM0** then you do not have to make any changes to the python scripts, because they are looking for **ACM0**.

Getting the software onto your RPi

1. Log into your Raspberry Pi4/5 using VNC.
2. Open a terminal window, type the following command and press enter

git clone https://github.com/UKradioastro/UKRAA_Magnetometer



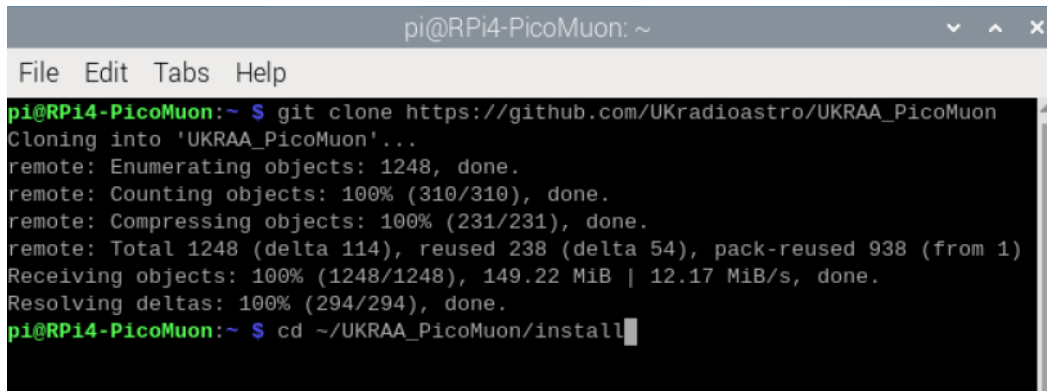
```
pi@RPi4-PicoMuon: ~
File Edit Tabs Help
pi@RPi4-PicoMuon:~ $ git clone https://github.com/UKradioastro/UKRAA_PicoMuon
```

This will download all of the code to the directory **UKRAA_Magnetometer** inside **/home/pi**

Install the software onto your RPi

1. Open a terminal window and type the following command and press **enter**

cd ~/UKRAA_Magnetometer/install

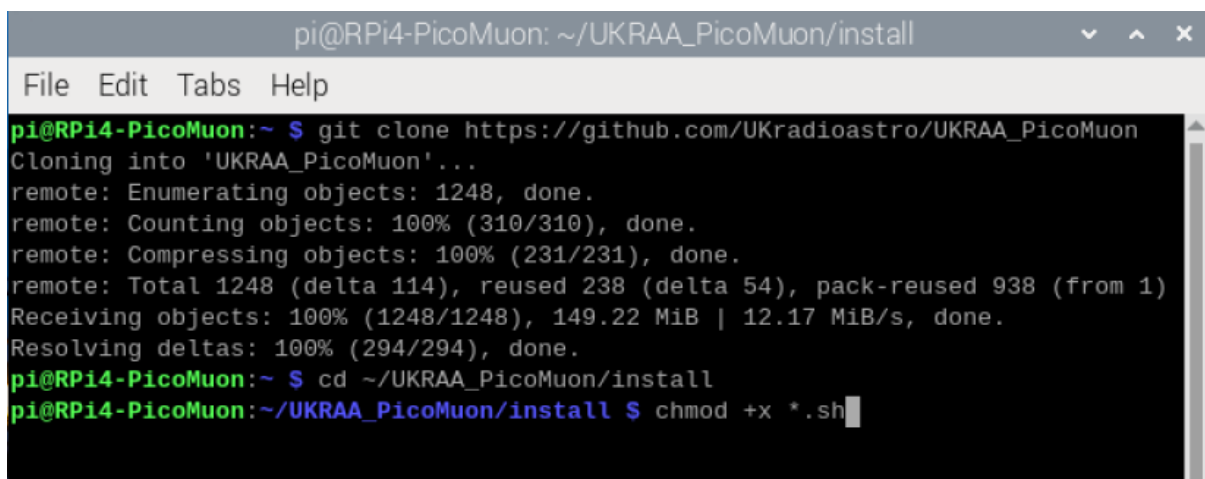
A terminal window titled 'pi@RPi4-PicoMuon: ~' with a menu bar (File, Edit, Tabs, Help). The terminal shows the execution of 'git clone https://github.com/UKradioastro/UKRAA_PicoMuon' and 'cd ~/UKRAA_PicoMuon/install'. The git clone output includes: 'Cloning into 'UKRAA_PicoMuon'...', 'remote: Enumerating objects: 1248, done.', 'remote: Counting objects: 100% (310/310), done.', 'remote: Compressing objects: 100% (231/231), done.', 'remote: Total 1248 (delta 114), reused 238 (delta 54), pack-reused 938 (from 1)', 'Receiving objects: 100% (1248/1248), 149.22 MiB | 12.17 MiB/s, done.', and 'Resolving deltas: 100% (294/294), done.' The prompt then changes to 'pi@RPi4-PicoMuon: ~ \$' after the cd command.

```
pi@RPi4-PicoMuon: ~ $ git clone https://github.com/UKradioastro/UKRAA_PicoMuon
Cloning into 'UKRAA_PicoMuon'...
remote: Enumerating objects: 1248, done.
remote: Counting objects: 100% (310/310), done.
remote: Compressing objects: 100% (231/231), done.
remote: Total 1248 (delta 114), reused 238 (delta 54), pack-reused 938 (from 1)
Receiving objects: 100% (1248/1248), 149.22 MiB | 12.17 MiB/s, done.
Resolving deltas: 100% (294/294), done.
pi@RPi4-PicoMuon: ~ $ cd ~/UKRAA_PicoMuon/install
```

This will take you to the **install** directory inside **/home/pi/UKRAA_Magnetometer**

2. Type the following command and press **enter**

chmod +x *.sh

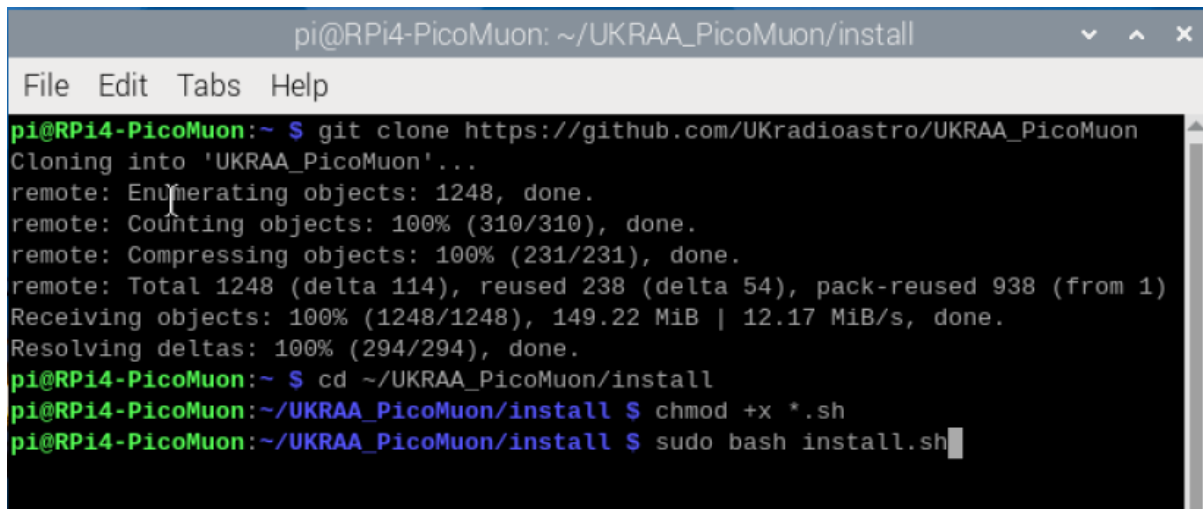
A terminal window titled 'pi@RPi4-PicoMuon: ~/UKRAA_PicoMuon/install' with a menu bar (File, Edit, Tabs, Help). The terminal shows the same git clone output as the previous screenshot, followed by the execution of 'cd ~/UKRAA_PicoMuon/install' and 'chmod +x *.sh'. The prompt changes to 'pi@RPi4-PicoMuon: ~/UKRAA_PicoMuon/install \$' after the chmod command.

```
pi@RPi4-PicoMuon: ~/UKRAA_PicoMuon/install $ git clone https://github.com/UKradioastro/UKRAA_PicoMuon
Cloning into 'UKRAA_PicoMuon'...
remote: Enumerating objects: 1248, done.
remote: Counting objects: 100% (310/310), done.
remote: Compressing objects: 100% (231/231), done.
remote: Total 1248 (delta 114), reused 238 (delta 54), pack-reused 938 (from 1)
Receiving objects: 100% (1248/1248), 149.22 MiB | 12.17 MiB/s, done.
Resolving deltas: 100% (294/294), done.
pi@RPi4-PicoMuon: ~/UKRAA_PicoMuon/install $ cd ~/UKRAA_PicoMuon/install
pi@RPi4-PicoMuon: ~/UKRAA_PicoMuon/install $ chmod +x *.sh
```

This will make the **install.sh** script executable.

3. Type the following command and press **enter**

sudo bash install.sh

A screenshot of a terminal window titled 'pi@RPi4-PicoMuon: ~/UKRAA_PicoMuon/install'. The terminal shows the execution of a git clone command to fetch the repository, followed by a cd command to navigate to the 'install' directory. Then, the user runs 'chmod +x *.sh' to make the script executable, and finally 'sudo bash install.sh' to run the installation script. The output of the git clone command shows progress for enumerating, counting, and compressing objects, and receiving the repository data.

```
pi@RPi4-PicoMuon: ~/UKRAA_PicoMuon/install
File Edit Tabs Help
pi@RPi4-PicoMuon:~ $ git clone https://github.com/UKradioastro/UKRAA_PicoMuon
Cloning into 'UKRAA_PicoMuon'...
remote: Enumerating objects: 1248, done.
remote: Counting objects: 100% (310/310), done.
remote: Compressing objects: 100% (231/231), done.
remote: Total 1248 (delta 114), reused 238 (delta 54), pack-reused 938 (from 1)
Receiving objects: 100% (1248/1248), 149.22 MiB | 12.17 MiB/s, done.
Resolving deltas: 100% (294/294), done.
pi@RPi4-PicoMuon:~ $ cd ~/UKRAA_PicoMuon/install
pi@RPi4-PicoMuon:~/UKRAA_PicoMuon/install $ chmod +x *.sh
pi@RPi4-PicoMuon:~/UKRAA_PicoMuon/install $ sudo bash install.sh
```

This will run the install script.

There will be occasions during the running of the install script that require you to make a keyboard entry.

When may be asked **Do you want to continue? [Y/n]** - type **Y** or **y** and press **enter**

That's it!

The code is now set up to run automatically; it will get the data from the PicoMagnetometer, process the data, plot the data and post the plots to your intranet web page.

What does the code do?

The code receives the event data from the UKRAA PicoMagnetometer via serial over the supplied USB cable and stores the event data to the raw data folder.

The raw data will be processed, via CRON, to get magnetic activity per minute, the variation of the X (N->S) and Y (E->W) magnetic field activity per hour and then plotted to relevant graphs.

Two plots will be created:

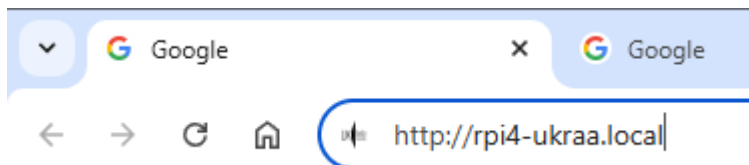
- Current activity plot
- Current X, Y and Z magnetic field variations

A simple web server and web page is set up on your RPi so that you can view your PicoMagnetometer's results on your desktop PC and/or smart phone when connected to your home network.

To access the webpage on your desktop PC or your smart phone...

1. Open your preferred web application (Safari, Chrome, Firefox, etc.).
2. In the search bar type the following and press enter

`http://rpi4-UKRAA.local`



This will take you to the web page for your magnetometer on your local intranet, displaying yesterday's events graphs.

NOTE: if you have a different **hostname** for your RPi, change the search bar entry to...

<http://hostname.local>

Where **hostname** is the hostname for your RPi setup.

Check GetDataRaw.py is running

1. To check the **status** of your service, type the following command and press enter.

sudo systemctl status PicoMagnetometer.service

2. To **start** your service, type the following command and press enter.

sudo systemctl start PicoMagnetometer.service

3. To **stop** your service, type the following command and press enter.

sudo systemctl stop PicoMagnetometer.service

4. To **enable** your service, type the following command and press enter.

sudo systemctl enable PicoMagnetometer.service

5. To **disable** your service, type the following command and press enter.

sudo systemctl disable PicoMagnetometer.service

License

MIT License

Copyright (c) 2024 UKRAA

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the **Software**), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED **AS IS**, WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.

Contact us

Please send an e-mail to Magnetometer@ukraa.com

